Lecture 10

Electoral Competition

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Remember (our simplified version of) Hotelling-Down

We zoomed into politicians' incentives

Assume away problem of how voters vote:

- We just assume they get utility from voting
- More reasonable: assume that the care about outcomes
- This might have consequences for candidates' choices

But we need, once again, to tool up

Quadratic Policy Preferences

n odd voters, set of possible policies is $\mathbb{R}$

Each voter has a favorite policy p_i: her ideal point

Voter i's preferences represented by $u_i(p) = -(p_i - p)^2$

These preferences are single-peaked

- Black's Theorem applies
- Median of p_is is the unique Condorcet Winner

Comparing Policies

Consider two policies: x and y, with $x > y$

Voter i prefers x to y if and only if $u_i(x) > u_i(y)$

$$-(p_i - x)^2 > -(p_i - y)^2,$$

$$2p_i(x - y) > x^2 - y^2 = (x + y)(x - y),$$

$$p_i > \frac{x + y}{2}$$

Voters who prefer the larger policy (x) are exactly those voters with ideal points greater than the midpoint of the two policies

Median Voter

Let p_{med} be median of the ideal points

Voter whose ideal point is p_{med} called the median voter

Majority Preference Equals Median Voter Preference

Assume $p_{\text{med}} > \frac{x+y}{2}$, so median voter prefers larger policy

$\frac{n-1}{2}$ voters have ideal points as great as p_{med}

All these voters prefer x to y

Thus majority prefer x to y

Majority preference is identical to preference of the median voter

This is stronger than Black's result (about Condorcet Winner)

A More General Model - 1

Does the MVT apply beyond unidimensional case?

Policy space is a subset of $\mathbb{R}^m$, so $\mathbf{p} \in \mathbb{R}^m$

Index the preferences by real numbers: $\Theta \subset \mathbb{R}$ is space of types

A voter is type $\theta \in \Theta$, has preferences given by $\succsim_\theta$ or $u(\cdot, \theta)$

- Two voters with same type have same preferences

Define θ_{med} to be the median of the types

Intermediate Preferences

A preference profile $\{\succsim_\theta\}_{\theta\in\Theta}$ satisfies Intermediate preferences if their utility can be written as

$$u(\mathbf{p},\theta) = J(\mathbf{p}) + K(\theta)H(\mathbf{p})$$

with $K(\cdot)$ monotonic

Theorem

Let $\{\succsim_\theta\}_{\theta\in\Theta}$ be a profile that satisfies intermediate preferences. Then a Condorcet winner exists and is given by the policy preferred by the median voter (θ_{med}).

A More General Model - 2

Does our stronger version of the MVT apply beyond quadratic case?

Policy space is a subset of $\mathbb{R}$

Index the preferences by real numbers: $\Theta \subset \mathbb{R}$ is space of types

A voter is type $\theta \in \Theta$, has preferences given by $\succsim_\theta$ or $u(\cdot, \theta)$

- Two voters with same type have same preferences

Define θ_{med} to be the median of the types

Monotonic Preferences

A preference profile $\{\succsim_\theta\}_{\theta\in\Theta}$ satisfies monotonicity if, for every pair of policies $p > p'$, there exists some some θ^* such that the set of voters who prefer p are exactly those with types greater than some θ^*

The type θ^* is often called a cutpoint

Median Voter Theorem

Theorem

Let $\{\succsim_\theta\}_{\theta\in\Theta}$ be a profile that satisfies monotonicity. Then a policy p is preferred by a majority to a policy p' if and only if p is preferred to p' by the median voter. That is, the majority preference relation c is identical to $\succsim_{\theta_{med}}$

Sufficiency

- Assume $p \succsim_{\theta_{med}} p'$
- By monotonicity, $\theta_{\text{med}} \geq \theta^*$
- By definition of median, majority of the voters have types greater than or equal to θ^*
- By monotonicity, this majority prefers p to p', so $p \succsim_M p'$.

Median Voter Theorem, cont'd

Theorem

Let $\{\succsim_\theta\}_{\theta\in\Theta}$ be a profile that satisfies monotonicity. Then a policy p is preferred by a majority to a policy p' if and only if p is preferred to p' by the median voter. That is, the majority preference relation ($\succsim_M$) is identical to $\succsim_{\theta_{med}}$

Necessity

- Assume $p \succsim_M p'$ (a majority of voters prefer p)
- By monotonicity, the set of such voters is exactly those with types greater than or equal to some θ^*
- By definition of the median, this must include θ_{med}
- By monotonicity, the median voter prefers p to p'

Increasing Differences

For $p > p'$, $u(p, \cdot) - u(p', \cdot)$ is the incremental return

Increasing differences: incremental return is increasing in type θ: for all $\theta > \theta'$

$$u(p, \theta) - u(p', \theta) > u(p, \theta') - u(p', \theta')$$

Recognizing Monotone Preferences

Need a way to recognize monotonicity

For any two policies $p > p'$, consider the maximization problem

$$\max_{x \in \{p, p'\}} u(x, \theta).$$

Call the solution $p^*(\theta)$

If u has increasing differences , then $p^*(\theta)$ is nondecreasing

p^* nondecreasing is exactly the monotonicity condition for the median voter theorem

Single Crossing

For $p > p'$, $u(p, \cdot) - u(p', \cdot)$ is the incremental return

Single crossing weakens increasing differences

- u satisfies the single crossing property if $p > p'$, $\theta > \theta'$, and $u(p, \theta') - u(p', \theta') \geq 0$, implies that $u(p, \theta) - u(p', \theta) \geq 0$.
- The incremental return changes sign only once

Another Monotonicity Theorem

Theorem

Assume that u satisfies the single crossing property and let $p^*(\theta) = \arg\max_{\{p,p'\}} u(p,\theta)$. Then p^* is nondecreasing in θ.

If the profile satisfies the single crossing property and all preferences are transitive [utility representation ensures it], then the majority preference is transitive, and there is a Condorcet winner.

- The corollary is immediate from the fact that $p^*(\theta)$ increasing implies monotone preferences

Why Generalize the MVT?

There are many substantive applications where preferences satisfy monotonicity but are not quadratic

Next week we will consider an example

A Simple Example

A continuum of citizens, indexed by $i \in [0, 1]$

Citizen i has income $w_i \geq 0$

$\overline{w}$ is the mean income and w_{med} is the median income

Society uses majority rule to choose linear income tax

If the tax system is (τ, T), a citizen with income w has after-tax income $(1 - \tau)w + T$

Government must balance its budget, so T equals average of tax payments $\tau\overline{w}$

Voter Preferences have Increasing Differences

Assume citizens maximize after-tax incomes

A citizen's utility over tax rates is

$$u(\tau, w_i) = (1-\tau)w_i + \tau\overline{w} = w_i + \tau(\overline{w} - w_i)$$

Checking for increasing differences

$$u_1(\tau, w_i) = \overline{w} - w_i$$

$$u_{12} = -1 < 0$$

So u has increasing differences in τ and $-w_i$

The Condorcet Winning Tax Rate

If citizen i greater than average income ($w_i > \overline{w}$), her utility is decreasing in the tax rate. Favorite tax rate is $\tau = 0$

If citizen i has less than the average income ($w_i < \overline{w}$), her favorite tax rate is $\tau = 1$

If median income is greater than average, then majority have favorite tax rate $\tau = 0$, while if the median income is less than the average, then majority have favorite tax rate $\tau = 1$

What's Wrong with this Model

Previous model predicts income tax rates will be either 0 or 1

Most developed countries have income tax rates that are strictly between 0 and 1

Can we modify model to capture this fact?

An obvious shortcoming of the model is that incomes are not responsive to taxes. But we believe that incomes decrease as taxes go up

Taxes and Labor Supply, Revisited

Each worker chooses her labor supply x_i to maximize $(1-\tau)w_i x_i + T + v(L - x_i)$

- L is the total time endowment
- τ is the income tax rate
- w_i is a wage rate
- T is transfer paid to every citizen

Government must balance its budget, so T is equal to the per-capita tax receipts

Can summarize policy with just one number, τ

Labor Market Behavior

Fix an arbitrary tax rate τ

The optimal labor market behavior for worker i, given τ, is

$$x^*(\tau, w_i) = \arg\max_x (1-\tau) w_i x + T(\tau) + v(L-x)$$

For future reference, notice that x^* satisfies the first-order condition

$$(1-\tau) w_i - v'(L - x^*) = 0$$

Also $x_1^*(\tau, w_i) < 0$ (just take cross-partial to see this)

Preferences over Tax Rates

A worker's indirect utility for a policy τ is her payoff under that policy, given that she will optimize in the labor market

$$V(\tau, w_i) = \max_x (1-\tau) w_i x + T(\tau) + v(L - x)$$

Substitute optimal labor market behavior to get

$$V(\tau, w_i) = (1-\tau) w_i x^*(\tau, w_i) + T(\tau) + v\left(L - x^*(\tau, w_i)\right)$$

Now check that the indirect utility function has increasing differences

Preferences over Tax Rates are Monotone

$$
\begin{aligned}
&V_2(\tau, w_i) \\
&\qquad = (1-\tau)x^*(\tau, w_i) + \left[(1-\tau)w_i - v'\left(L - x^*(\tau, w_i)\right)\right] x_2^*(\tau, w \\
&\qquad = (1-\tau)x^*(\tau, w_i).
\end{aligned}
$$

- The first line is messy but direct
- Second line follows from worker's first order condition, which says the part is square parentheses is zero

$$V_{12}(\tau, w_i) = -x^*(\tau, w_i) + (1-\tau)x_1^*(\tau, w_i) < 0,$$

- Inequality follows from our earlier analysis
- Indirect utilities satisfy monotonicity, so the median wage rate worker's favorite tax rate is the Condorcet winner

The Condorcet Winning Tax Rate

Median voter's indirect utility is

$$V(\tau, w_{\mathrm{m}}) = (1-\tau)w_{\mathrm{m}}x^*(\tau, w_{\mathrm{m}}) + T(\tau) + v\left(L - x^*(\tau, w_{\mathrm{m}})\right)$$

Median voter's favorite tax rate satisfies the first-order condition

$$\begin{gathered} -w_{\mathrm{m}}x^*(\tau_{\mathrm{m}}^*, w_{\mathrm{m}}) + T'(\tau_{\mathrm{m}}^*) \\ + \left[(1-\tau)w_{\mathrm{m}} - v'\left(L - x^*(\tau, w_{\mathrm{m}})\right)\right] x_1^*(\tau_{\mathrm{m}}^*, w_{\mathrm{m}}) = 0 \end{gathered}$$

Using first-order condition from labor supply, rewrite

$$-w_{\mathrm{m}}x^*(\tau_{\mathrm{m}}^*, w_{\mathrm{m}}) + T'(\tau_{\mathrm{m}}^*) = 0$$

We can learn more if we are more explicit about the function T

Condorcet Winning Tax Rate, cont'd

$$-w_{\mathrm{m}} x^*(\tau^*_{\mathrm{m}}, w_{\mathrm{m}}) + T'(\tau^*_{\mathrm{m}}) = 0$$

Let f be the density of wages in the population, then we can write $T(\tau) = \tau \int \tilde{w} x^*(\tilde{w}, \tau) f(\tilde{w})\, d\tilde{w}$ and

$$T'(\tau) = \int \tilde{w} x^*(\tau, \tilde{w}) f(\tilde{w})\, d\tilde{w} + \tau \int \tilde{w} x_1^*(\tau, \tilde{w}) f(\tilde{w})\, d\tilde{w}$$

Now rewrite the median voter's first-order condition as

$$-w_{\mathrm{m}} x^*(\tau^*_{\mathrm{m}}, w_{\mathrm{m}}) + \int \tilde{w} x^*(\tau^*_{\mathrm{m}}, \tilde{w}) f(\tilde{w})\, d\tilde{w}$$

$$= -\tau^*_{\mathrm{m}} \int \tilde{w} x_1^*(\tau^*_{\mathrm{m}}, \tilde{w}) f(\tilde{w})\, d\tilde{w}$$

Condorcet Winning Tax Rate, cont'd

$$-w_{\mathrm{m}} x^*(\tau^*_{\mathrm{m}}, w_{\mathrm{m}}) + \int \tilde{w} x^*(\tau^*_{\mathrm{m}}, \tilde{w}) f(\tilde{w})\, d\tilde{w}$$

$$= -\tau^*_{\mathrm{m}} \int \tilde{w} x^*_1(\tau^*_{\mathrm{m}}, \tilde{w}) f(\tilde{w})\, d\tilde{w}$$

Since $x^*_1 < 0$, the function $K(\tau) = -\int \tilde{w} x^*_1(\tau, \tilde{w}) f(\tilde{w})\, d\tilde{w}$ is positive, and we can write the equilibrium tax as

$$\tau^*_{\mathrm{m}} = \frac{\overline{y} - y_{\mathrm{m}}}{K(\tau^*_{\mathrm{m}})},$$

where y is income

Intuitively, K represents the dead weigh loss of taxation. The median voter now must trade off the gain from redistribution against this dead weigh loss

Empirical Evidence

Prediction: Taxes are increasing in the difference between the mean and median incomes

This prediction is not very successful in the data (see pp. 121–123 of Persson and Tabellini's *Political Economics* for discussion and references)

Much more redistribution in Europe than in the US

	US	**UK**	**Europe**
Size of Government	29.9	37.3	44.9
Spending on Social Programs	14.6	24.7	25.5
Tax Revenue	31.0	38.7	46.7

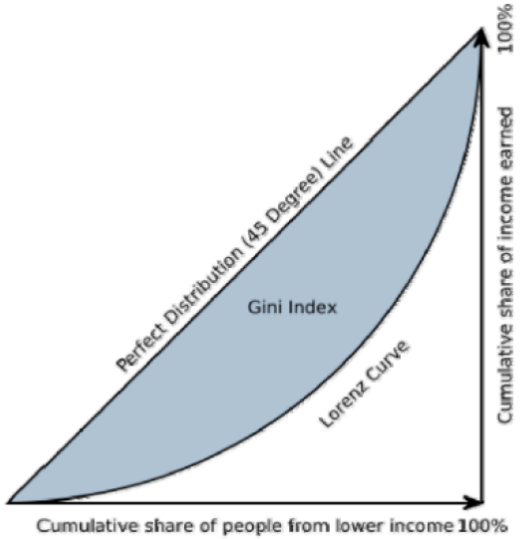
Redistribution in the US vs. Europe. All figures are as a percentage of GDP, and are taken from Alesina and Glaeser.

Empirical Evidence, cont'd

Also, taxes are more progressive in Europe

> [M]arginal tax rates in the United States are higher than in Europe for low levels of income (up to about 50 percent of the average worker's wage) and lower for higher levels of income. Also, the difference between the United States and Europe becomes larger in absolute value as income rises. [Alesina and Glaeser, p. 37]

Empirical Evidence, cont'd



45° line is perfect equality. Curved line reflects an empirical distribution

Gini Coefficient is the the distance from perfect equality. Smaller Gini, less inequality

US Gini is 38.5 vs. 29.1 in Europe (Alesina and Angeletos). Our simple model cannot explain the variation

One More Really Useful Tool

- Several times in the income tax model, used worker's first-order condition for optimal labor supply to simplify derivative of voter's indirect utility function over taxes
- This trick is useful all over formal theory
- Let $a^*(\theta)$ be a solution to our standard optimization problem

$$\max_{a \in A} f(a, \theta),$$

- For simplicity, assume that a^* is a singleton for every θ.
- Let $V(\theta)$ be the **maximum value function**

$$V(\theta) = f\left(a^*(\theta), \theta\right)$$

The Envelope Theorem

Theorem

Assume that f is differentiable in both of its arguments. Then V is differentiable with derivative $V'(\theta) = f_2\left(a^*(\theta), \theta\right)$.

- Assume a^* is differentiable and interior for all θ. Implies it satisfies the first-order condition

$$f_1\left(a^*(\theta), \theta\right) = 0$$

- By the chain rule, the derivative of the value function is

$$V'(\theta) = f_1\left(a^*(\theta), \theta\right) \frac{da^*}{d\theta}(\theta) + f_2\left(a^*(\theta), \theta\right)$$

- By the first-order condition, this can be written

$$V'(\theta) = f_2\left(a^*(\theta), \theta\right)$$

How General?

- Envelope Theorem does not rely on first-order condition
- Assume $A = \{0, 1\}$ and $f(a, \theta) = -(a - \theta)^2$.

$$a^*(\theta) = \begin{cases} 0 & \text{if } \theta \in [0, 1/2] \\ 1 & \text{if } \theta \in [1/2, 1] \end{cases}$$

- The value function is

$$V(\theta) = \begin{cases} -\theta^2 & \text{if } \theta \leq \frac{1}{2} \\ -(1-\theta)^2 & \text{if } \theta > \frac{1}{2} \end{cases}.$$

- If $\theta \in [0, 1/2)$ the derivative is $V'(\theta) = -2\theta$, and if $\theta \in (1/2, 1]$ the derivative is $V'(\theta) = 2(1 - \theta)$.
- So the Envelope Theorem holds at every θ where there is a unique solution

Convergence

Another implication of the model is that two policy-motivated candidates (R and L) should converge to the median's preferred taxation

Also not realistic

Solution: assume probabilistic voting: each voter votes for R iff

$$V(\tau^R, w_i) \geq V(\tau^L, w_i) + \epsilon$$

where ϵ is the realization of a random variable

Preference shock captures other idiosyncratic factors and other policy dimensions affecting voting choice

$$Pr(\text{R wins}) = p_R = F_\epsilon[V(\tau_R, w_i) - V(\tau_L, w_i)]$$

Example

Suppose that J has payoff $-(\tau - \hat{\tau}_J)^2$, where $\hat{\tau}_R < \tau^*_{\mathrm{m}} < \hat{\tau}_L$

Each party maximizes $-p_J(\tau_J - \hat{\tau}_J)^2 - (1 - p_J)(\tau_{-J} - \hat{\tau}_J)^2$

Assume ϵ is drawn from a uniform with support $\left[-\frac{1}{2\psi}, \frac{1}{2\psi}\right]$

Then $p_J = \frac{1}{2} + \psi[V(\tau_R, w_{\mathrm{m}}) - V(\tau_L, w_{\mathrm{m}})]$

Candidate's FONC given by

$$\frac{\partial V(\tau, w_{\mathrm{m}})}{\partial \tau}[(\tau_{-J} - \hat{\tau}_J)^2 - (\tau - \hat{\tau}_J)^2] - 2p_J(\tau - \hat{\tau}_J)$$

which implies partial convergence:

$$\hat{\tau}_R < \tau_R < \tau^*_{\mathrm{m}} < \tau_L < \hat{\tau}_L$$